Towards a new effective representation of polynomial systems of equations using Gröbner bases

We are seeking an excellent candidate for a 5 to 6-month research internship at the interface of computational mathematics and geometry, with the goal of opening the door towards developing new general-purpose tools in computer algebra.

The internship will be hosted by the PolSys team (Polynomial Systems) at LIP6, a joint research unit of Sorbonne Université and CNRS.

Key words: Computer algebra & symbolic computation, Polynomial system solving, algebraic methods, Gröbner bases

Gröbner bases and polynomial system solving

Polynomial systems of equations are able to model a wide variety of non-linear problems in the sciences, for example in robotics [11, 20], in chemistry [9, 6], in cryptography [13, 2], and in optimization [7, 1]. *Gröbner bases*, introduced by Buchberger in the 1960's [4], are the most widely used tool for the symbolic algorithmic treatment of polynomial systems. They form a central component of all modern computer algebra systems in this regard, such as `SageMath` and `Maple`.

Following the view of Lazard [14], Gröbner bases can be understood as a form of *linearization* of a polynomial system: Out of a polynomial system of equations $F = 0$ one constructs several so-called *Macaulay matrices* [16], each associated to a degree d. These matrices are then echelonized (using Gaussian elimination), up to a certain degree D, and the rows of these echelonized matrices form the desired Gröbner basis. The central challenge in the computation of Gröbner bases comes from the fact that these Macaulay matrices rapidly grow in size with growing d and growing number of variables and that, moreover, the maximal degree D up to which these matrices need to be constructed is in the worst case doubly exponential in the number of variables of F [17]. This doubly exponential worst case bound stands in contrast to the good practical behavior of Gröbner bases. This good practical behavior is partially visible in the fact that for systems F with *finitely many solutions*, the maximum degree D becomes only *linear* in the number of variables and the degrees of the equations F [14]. As a consequence, despite the fact that symbolic techniques are known which have a better controlled worst case complexity [10, 15], Gröbner bases remain the practical tool of choice for algorithmically treating polynomial systems.

However, several classes of polynomial systems of interest remain out of reach of a resolution via Gröbner basis computations. To give a prominent example, *matrix multiplication algorithms* for matrices of fixed size, constituting a currently highly active research field [12, 8] can be encoded as solutions of polynomial equations [3]. The number of involved variables and the *high dimension* of their solution sets, however, puts these systems out of reach of state-of-the-art symbolic techniques.

With this in mind, an the central question raised for this internship is whether one can design a new class of Gröbner basis algorithms that reduce computations with arbitrary polynomial systems to computations with polynomial systems with finitely many solutions, in the hopes of obtaining better worst-case complexities than plain Gröbner basis computations and in further improving their practical usability.

Goal and methodology of the internship

In this internship, we will take some first steps in answering the question above as follows:

Witness sets. First, suppose that the solution set $\{F = 0\}$ of a given polynomial system F is *equidimensional* of positive dimension δ. This means essentially that $\{F = 0\}$ does not consist of finitely many isolated points (which corresponds to the zero-dimensional case) and does not have components of different dimension. For example, $\{F = 0\}$ could consist of the union of two surfaces (so $\delta = 2$) but not of the union of a curve and a surface, i.e. a one- and two-dimensional component. It turns out that such a polynomial system F can then be *probabilistically* represented by a so-called *witness set* W: This is a finite point sample on $\{F = 0\}$ constructed by adjoining δ *random* linear equations to F, so W is a *zero-dimensional* representation of F, see also Figure 1. Such witness sets form one of the backbones of the *numerical* treatment of polynomial systems [19] and have been shown to form a desirable representation of F in symbolic computation from a complexity-theoretic point of view in the above-mentioned [10, 15]. More explicitly, the main objective of this internship is to take some first steps in combining such a witness set representation with Gröbner basis computations.

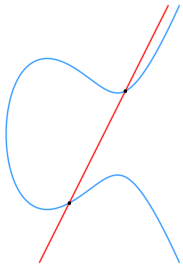


Figure 1: In cutting the curve $y^2 - x^3 + 2x - 2 = 0$ (in blue) with one linear constraint (in red), one obtains a witness set for X (in black).

Combining witness sets with Gröbner bases. The main difficulty in achieving this goal is that only *equidimensional* solution sets can be represented using a witness set. This will require us to combine a witness set based representation with an *equidimensional decomposition algorithm*, i.e. an algorithm that partitions $\{F = 0\}$ dimension-by-dimension. For this we will use an algorithm for equidimensional decomposition designed by the envisioned supervisor of this internship and given in [18]. As many other algorithms for equidimensional decomposition, it relies on the following algorithm pattern: Given the set $\{F = 0\}$, first use a Gröbner basis computation to find a polynomial g such that

$$\{F = 0\} = \{F = g = 0\} \cup \{F = 0, g \neq 0\}$$

is a non-trivial decomposition, i.e. separates parts of different dimension of $\{F = 0\}$. Next, treat recursively the two new systems $F = g = 0$ and $F = 0, g \neq 0$ in the same manner until equidimensionality is detected. A major difficulty in this algorithmic pattern is that the new system $F = g = 0$ includes certain "redundancies" which can cause a massive blow up in the number of treated polynomial systems further down the line. Because of this, [18] gives involved and expensive procedures for removing these redundancies which become the main bottleneck of the computation. However, adding linear constraints to $F = g = 0$ turns out to also remove some of these redundancies, without the need for additional computations. Therefore, combining a witness set based representation with the algorithm in [18] is expected to at the same time simplify it and improve its practical behavior.

Organization of the internship

Workplan of the internship. We will realize the scientific goal of this internship, i.e. combining the algorithm of [18] with witness sets, along the following work plan.

- Study the basic theory of the algorithmic treatment of polynomial systems of equations using Gröbner bases, e.g. along [5], as well as the decomposition algorithm of [18].
- Design new decomposition procedures for [18], combining it with a witness-set based representation,
- Start to study complexity-theoretic questions related to the algorithm designed in the second step by using known complexity results for Gröbner basis computations in dimension zero [14].

- Supply a prototype implementation of the algorithm obtained in the second step by using the `julia` package AlgebraicSolving.jl which is maintained in part by PolSys and already contains an implementation of [18].

Scientific environment. This internship will be supervised by Rafael Mohr. It will take place at Sorbonne Université, in the computer science lab LIP6, which is located on the Pierre-et-Marie-Curie campus, at the heart of Paris. The intern will be welcomed in the PolSys team, which develops and implements fast computer algebra algorithms, for polynomial system solving and their applications, such as coding theory, combinatorics, cryptography, and robotics. The intern will work in a kind and international environment gathering PhD students and Post-Docs representing several nationalities, and animated with several working groups and a monthly seminar. All computing facilities will be provided.

Weekly meetings with the supervisor will be organized.

Prerequisites. We are looking for exceptional candidates who are excited about computer algebra, algorithmic design in mathematics and software implementations. We expect the potential candidate to be able to communicate and write in English.

We expect curiosity and openness, strong interpersonal and communicative skills, and willingness to dive into new topics, including the design of powerful mathematical algorithms and their implementation.

How to apply. Applicants should send a full CV, a letter of motivation (in English or French) and the grades obtained during the last two years to

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